Appointment Reminder Service

PRD-APT-034 | Product Requirements Document

FICTIONAL VALIDATION DOCUMENT - contains no real customer or production data

Status	Draft	**Owner**	Digital Operations
Version	1.0	**Updated**	15 Oct 2026
Audience	Product, QA, Engineering	**Classification**	Synthetic / Non-production

1. Objective

Send appointment reminders through email and SMS, while preventing duplicate notifications and honoring each recipient's communication preferences. This specification uses invented clinics, identifiers, and data solely for parser validation.

2. Requirement catalogue

ID	Capability	Rule / acceptance criteria	Priority
APT-01	Schedule reminder	Create reminders at 24 hours and 2 hours before the appointment start time.	Must
APT-02	Timezone handling	Use the clinic timezone when computing send windows; persist timestamps in UTC.	Must
APT-03	Preference suppression	Skip SMS when SMS consent is absent; do not substitute another channel without consent.	Must
APT-04	Deduplicate	A patient, appointment, channel, and reminder offset combination may be sent at most once.	Must
APT-05	Reschedule	Cancel unsent reminders for the previous appointment time and create reminders for the new time.	Must
APT-06	Delivery status	Expose QUEUED, SENT, DELIVERED, FAILED, and SUPPRESSED to internal operators.	Should
APT-07	Retry	Retry transient errors after 2, 10, and 30 minutes; do not retry permanent provider rejections.	Must
APT-08	Quiet hours	For SMS, delay non-urgent sends that fall between 21:00 and 08:00 local time to 08:05.	Should
APT-09	Audit	Keep a 180-day event history for troubleshooting and operational	Should

ID	Capability	Rule / acceptance criteria	Priority
		reporting.	
APT-10	Template variables	Missing appointment date or clinic name blocks delivery and raises TEMPLATE_DATA_MISSING.	Must
APT-11	Operator replay	Permit one manual replay after an operator confirms consent and recipient data.	Could
APT-12	Metrics	Publish queued count, delivery latency, provider failure rate, and suppression count.	Should

3. Message delivery mapping

Provider code	Internal result	Definition	Required behavior
00	ACCEPTED	Provider accepted the message for delivery	Store provider message ID
10	DELIVERED	Provider reports delivery to recipient device or mailbox	Mark delivery complete
20	TEMPORARY_ERROR	Provider timeout or rate limit	Retry according to backoff schedule
30	PERMANENT_ERROR	Invalid destination or blocked recipient	Stop retries and alert operations
40	CONSENT_MISSING	Required channel consent is absent	Suppress without provider call
50	TEMPLATE_DATA_MISSING	Required template variable is unavailable	Suppress and create an operational alert
60	DUPLICATE	A matching reminder event already exists	Return original event identifier

4. Data dictionary

Field	Type	Nullability	Definition
appointment_id	VARCHAR(24)	Required	Stable appointment identifier; not a patient identifier.
scheduled_start_utc	TIMESTAMP	Required	ISO timestamp normalized to UTC before persistence.
clinic_timezone	VARCHAR(40)	Required	IANA timezone name such as Asia/Jakarta.
reminder_offset_minutes	INTEGER	Required	Allowed values: 1440 or 120.
channel	VARCHAR(8)	Required	EMAIL or SMS.
consent_version	VARCHAR(16)	Conditional	Required when channel is SMS.
provider_message_id	VARCHAR(64)	Optional	Assigned after provider

Field	Type	Nullability	Definition
			acceptance.
attempt_count	SMALLINT	Required	Starts at 0; maximum automatic attempts is 4.
last_error_code	VARCHAR(32)	Optional	Null when no delivery error has occurred.

5. Operational sequence

1. A scheduler selects reminders whose send_at is due.
2. The preference service verifies channel consent and quiet-hour rules.
3. The delivery worker obtains a deduplication lock before sending.
4. The provider response is mapped to an internal result code.
5. Metrics and audit events are recorded whether delivery succeeds or fails.

6. Example-only illustration

The following sentence is illustrative text, not a requirement: “Your appointment is tomorrow at 10:30.” Example-only content must not override the explicit rules above.